

ECON 100B: Macroeconomics

Class 14: The Short Run (Jones Chapter 9)

Ryan D. Edwards

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Agenda

- Announcements
- Current events
- Jones Chapter 9: The Short Run

Announcements

- Take-home Midterm Essay released 3/8, due 3/20
 - Scores will be high
 - You can use many tools including AI, but essays written solely by AI typically lose points and break the rules. Up to 10% of ECON 157 for example. AI is not a student in this classroom and won't know the best answers
- Spring Break 3/23-3/27

Current events

- February 2026 CPI inflation report from BLS was released Wednesday, March 11
- Lahart (3/11/26) “[Inflation Holds Steady, but Iran War Threatens to Boost Prices](#)”
 - Overall CPI inflation up 2.4% from February 2025
 - Core CPI inflation up 2.5% from February 2025
 - On Friday, BEA will release PCE inflation, expected to be higher because of trends in the components that differ between them

The picture shows why core inflation: it's more stable

Consumer-price index, change from a year earlier



Note: Data for October 2025 is unavailable. Core excludes food and energy prices.

Source: Labor Department

Where we are and where we're headed

- Chapters 1-8 are about the long run
 - (Supply) Labor, capital, total factor productivity create **potential output** $\bar{Y}$
 - Social trends and human capital accumulation create a labor force
 - Given $\bar{Y}$, money and credit create a long-run rate of inflation, $\bar{\pi}$, because $MV = PY$

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 - Given $\bar{Y}$, money and credit create a long-run rate of inflation, $\bar{\pi}$, because $MV = PY$
- Chapters 9-20 are (mostly) about the short run
 - (Demand) Expenditures can deviate from **potential output** $\bar{Y}$

$$Y = C + I + G + NX \lesseqgtr \bar{Y}$$

- A big reason is because **investment** responds to the real interest rate. Another is because all components respond to shocks. Those get amplified when consumption responds to current income
- When $Y > \bar{Y}$, the economy is booming, and $\tilde{Y} = \frac{Y - \bar{Y}}{\bar{Y}} > 0$ and the **Phillips Curve** shows us $\Delta\pi > 0$. When the economy is slumping: the reverse

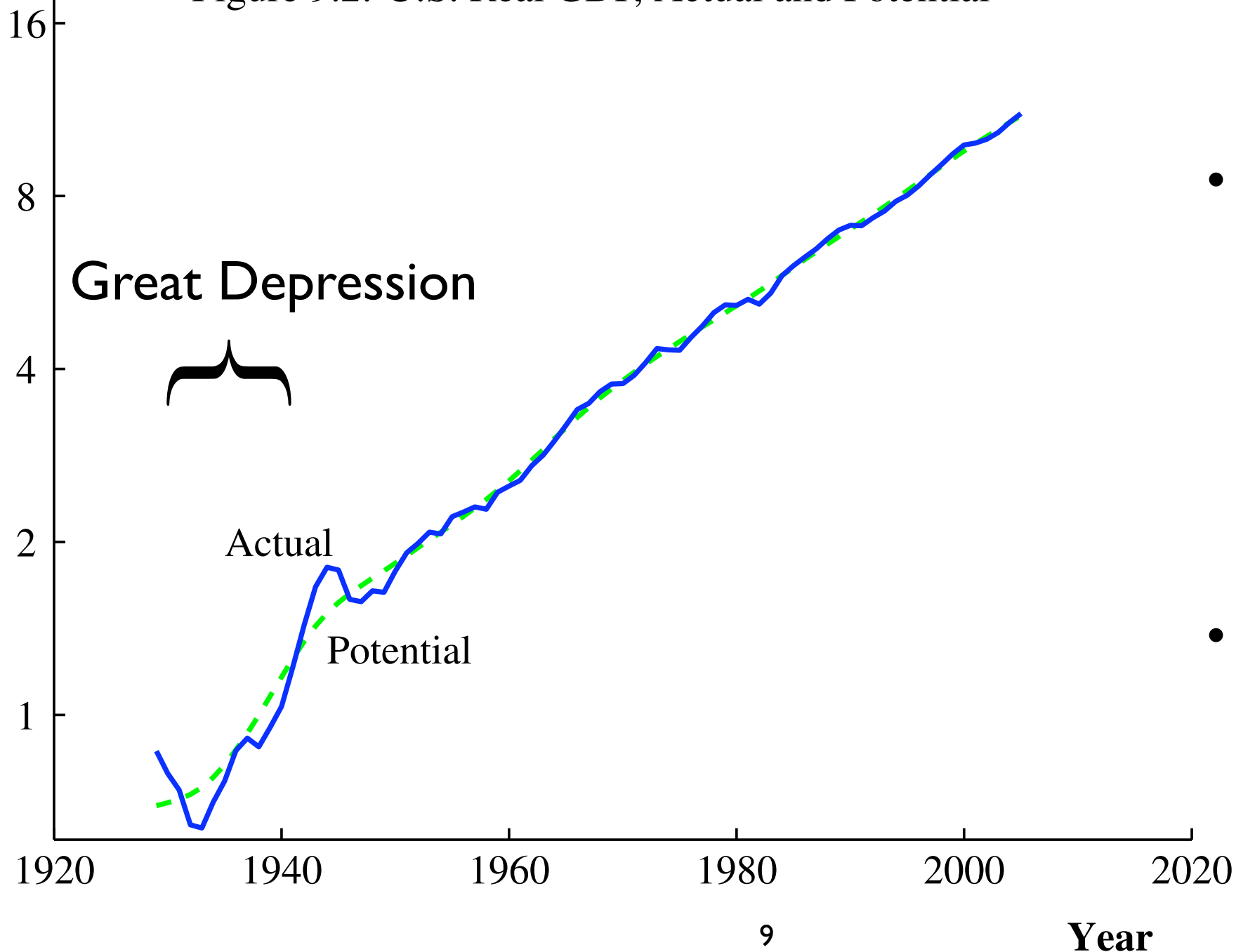
The Economy in the Short Run

- The growth chapters told us about well-being in the long run
- But there are many interesting short-run dynamics
- The **gap** between current GDP and *potential* GDP is a key measure of performance
 - “Current GDP” = what or how the economy is currently doing
 - “Potential GDP” = the economy’s potential, what it could be doing
- When the gap is negative, GDP and employment both fall
- But at least *inflation declines* when the gap is negative!

There are *wiggles* around the trend in GDP

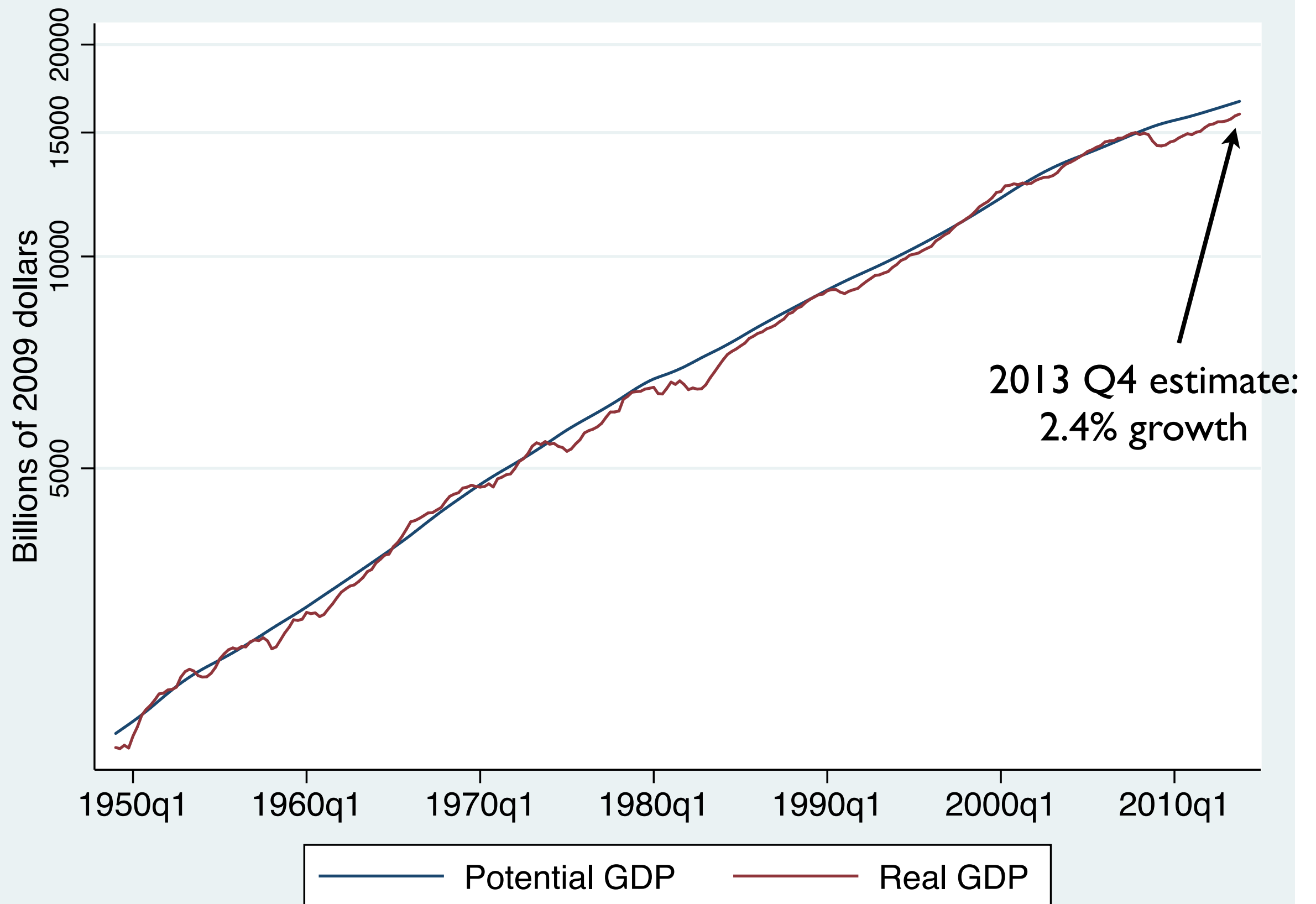
(trillions of chained
2000 dollars, ratio scale)

Figure 9.2: U.S. Real GDP, Actual and Potential

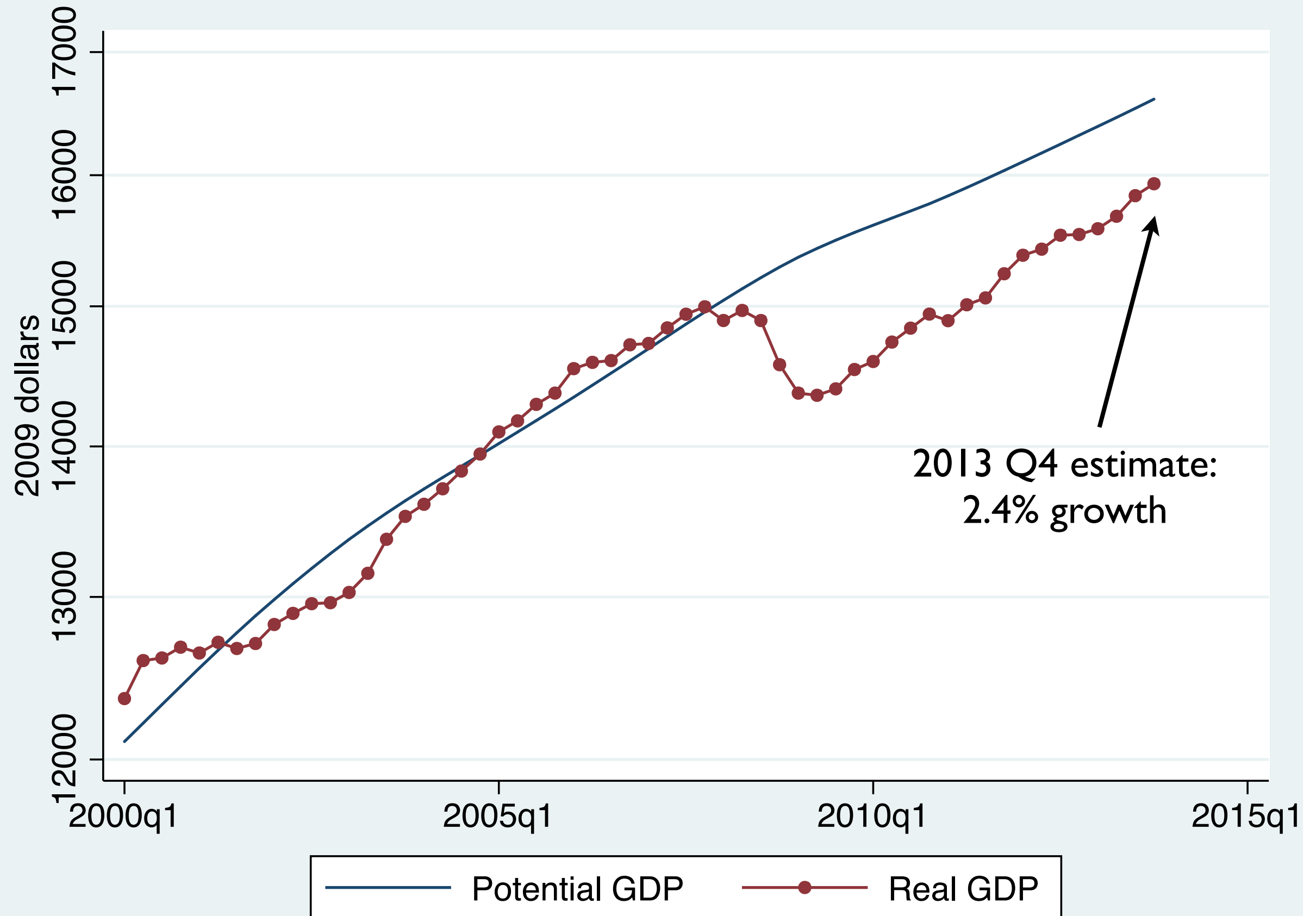


- So far, we've only talked about the trend, but the wiggles are important
- The biggest wiggle was the Great Depression, which was so traumatic that economists developed national income accounting
- It's hard to see the other wiggles, but they are there, just smaller

An updated graph with data from 1949 to 2013 Q4



Quarterly GDP data during the long recovery



Overview of the course

- Chapter 2: Actual GDP is always equal to spending:

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

- Chapters 3–6: In the long run, growth in **potential GDP** is determined by growth in technology (ideas), capital, and labor:

$$\bar{Y}_t = A_t K_t^{1/3} L_t^{2/3}$$

- Chapters 7–8: There is also inflation (π) and unemployment
- Chapters 9–20: In the short run, spending can exceed or fail to meet potential — due to changes in demand or in interest rates — with consequences for unemployment and inflation
- In other words, Y might not always equal $\bar{Y}$

Potential GDP in the long run

$$\bar{Y}_t = A_t K_t^{1/3} L_t^{2/3}$$

- In the first part of the course, we learned how “potential GDP” is obtained through:

Population growth at $g_L = 1.5\%$ we assume it just happens

Growth in ideas $g_A = 1.3\%$ because of R&D (Romer model)

Growth in capital at $g_K = 3.5\%$ because of saving (Solow Model)

- Growth in potential output is just $g_Y = g_A + 1/3 g_K + 2/3 g_L = \mathbf{3.5\%}$
- That could change if there were a change in g_L or g_A

To help tell stories, we break down GDP into long-run and short-run components

Current output we observe Y_t looks like a Long-run trend $\bar{Y}_t$ & short-run fluctuation ?

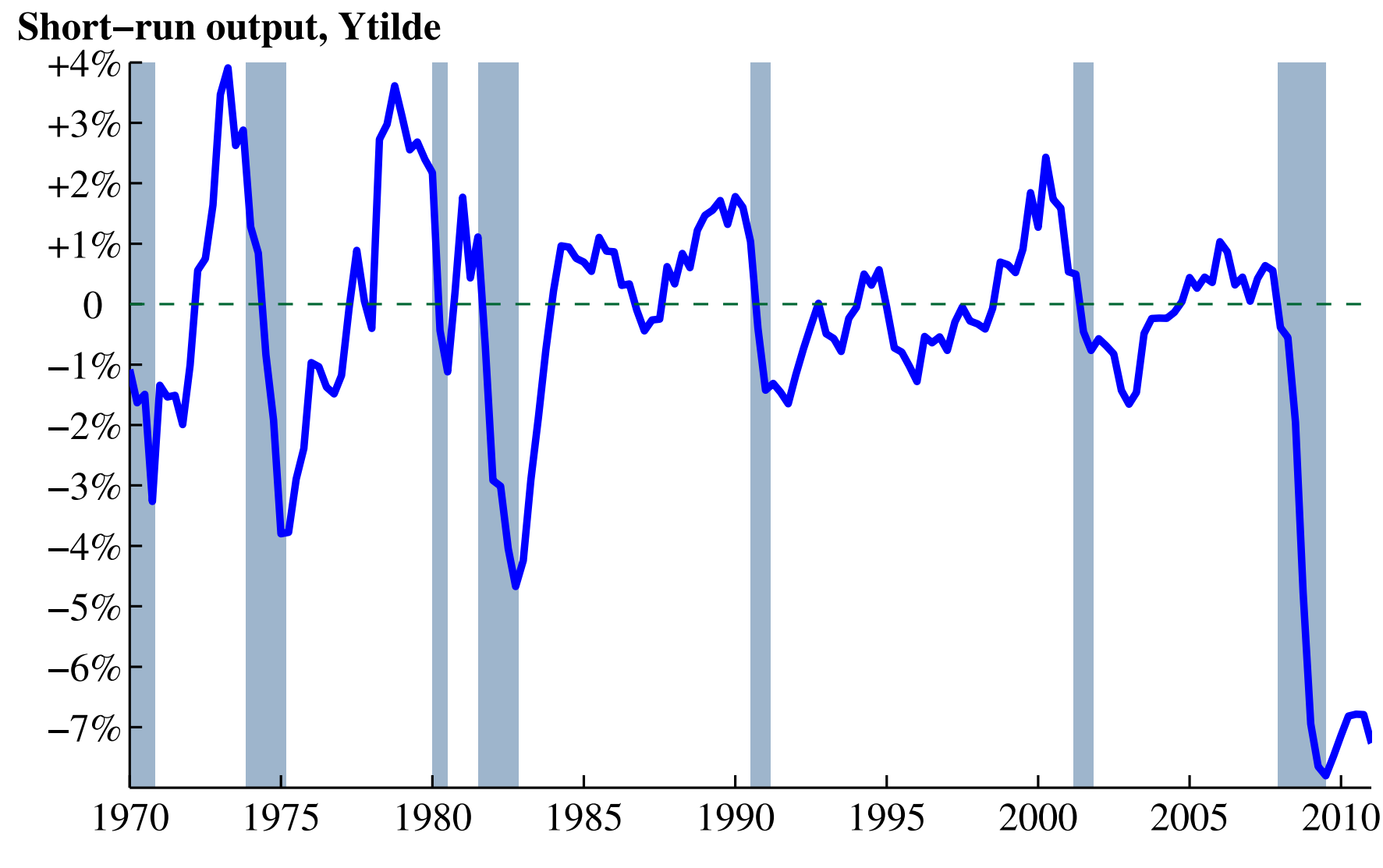
We're going to think in terms of *percentage deviations* from trend:

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}, \quad \tilde{Y}_t \text{ is "short-run output"}$$

In words: short-run output pulls us a percentage above or below trend.
Why? The size of fluctuations over time fits this pattern

Percentage deviations from trend remained within about a $\pm 4\%$ band ... until the Great Recession

Fig 9.3: U.S. Economic Fluctuations



Source: Federal Reserve Economic Data

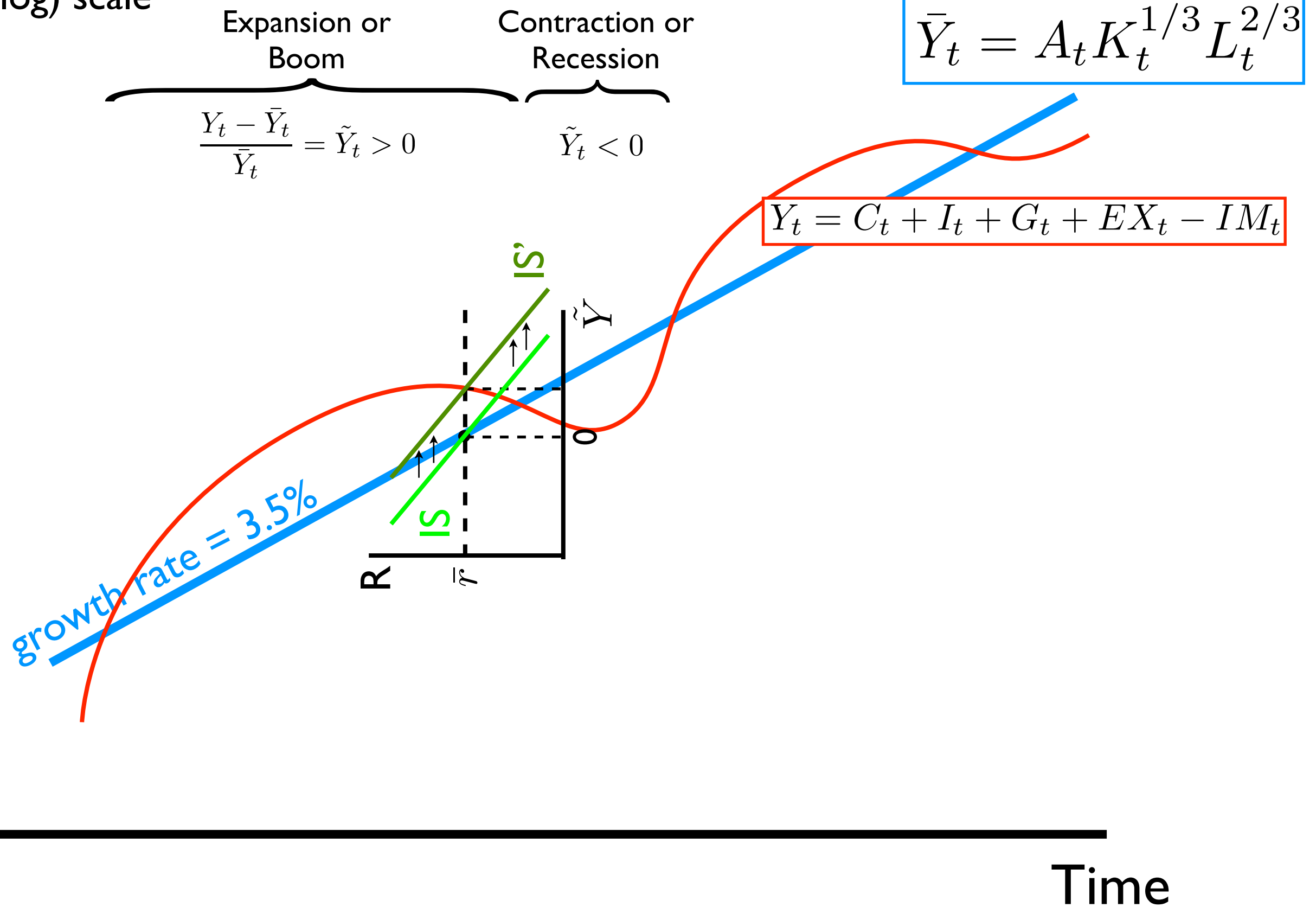
- The blue bars indicate periods specified as *recessions* by the National Bureau of Economic Research
- What kind of **costs** do these swings represent?
- A recession might be a 2-year swing that loses us about 6% of GDP relative to potential
- GDP \$12 trillion $\times$ 6% = \$720 billion, then $\div$ 300 million people $\Rightarrow$ \$2,400 per person

If it's helpful, here's a graphical view

It shows how the short run model
“fits inside” the long run model

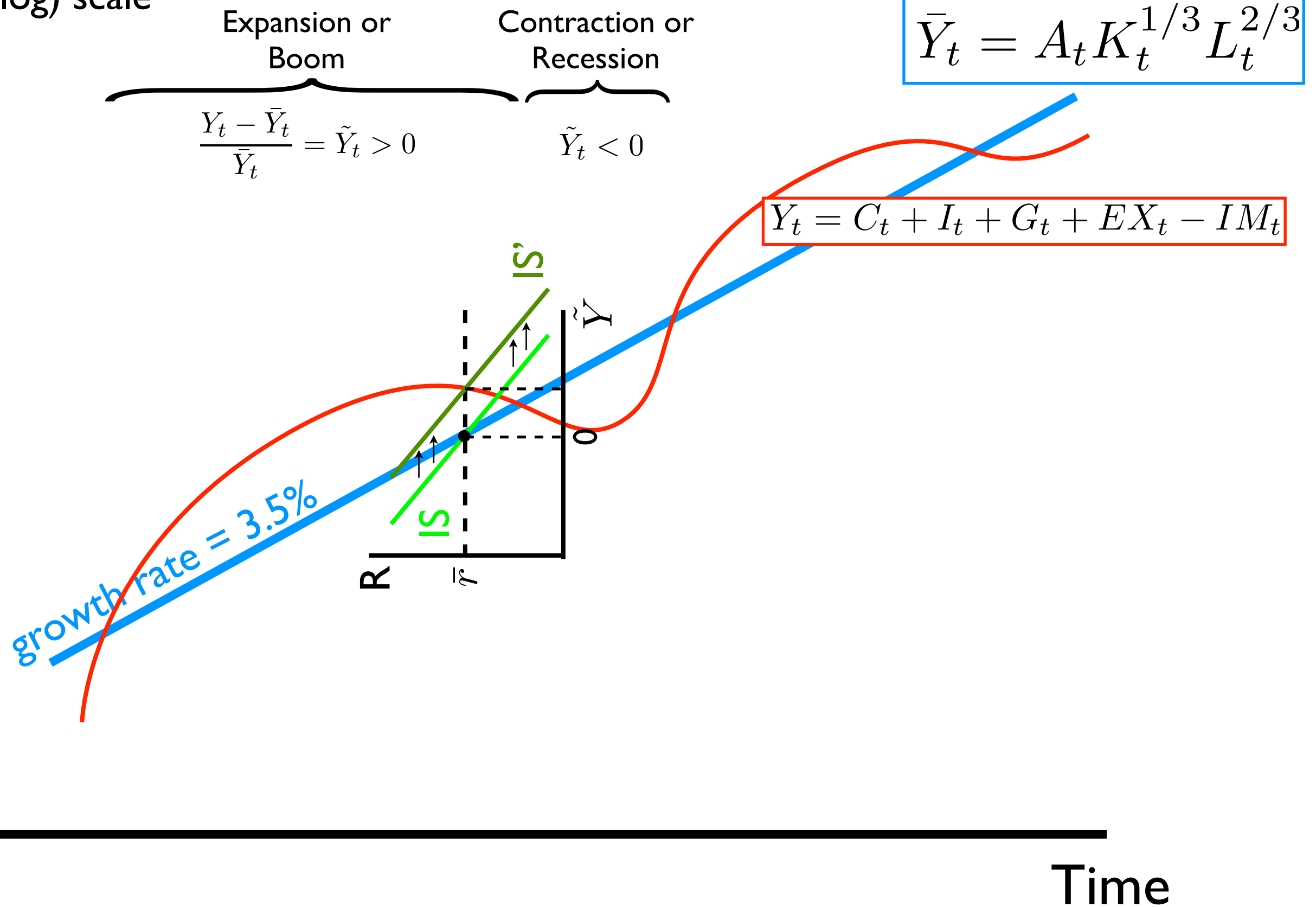
GDP

ratio (log) scale



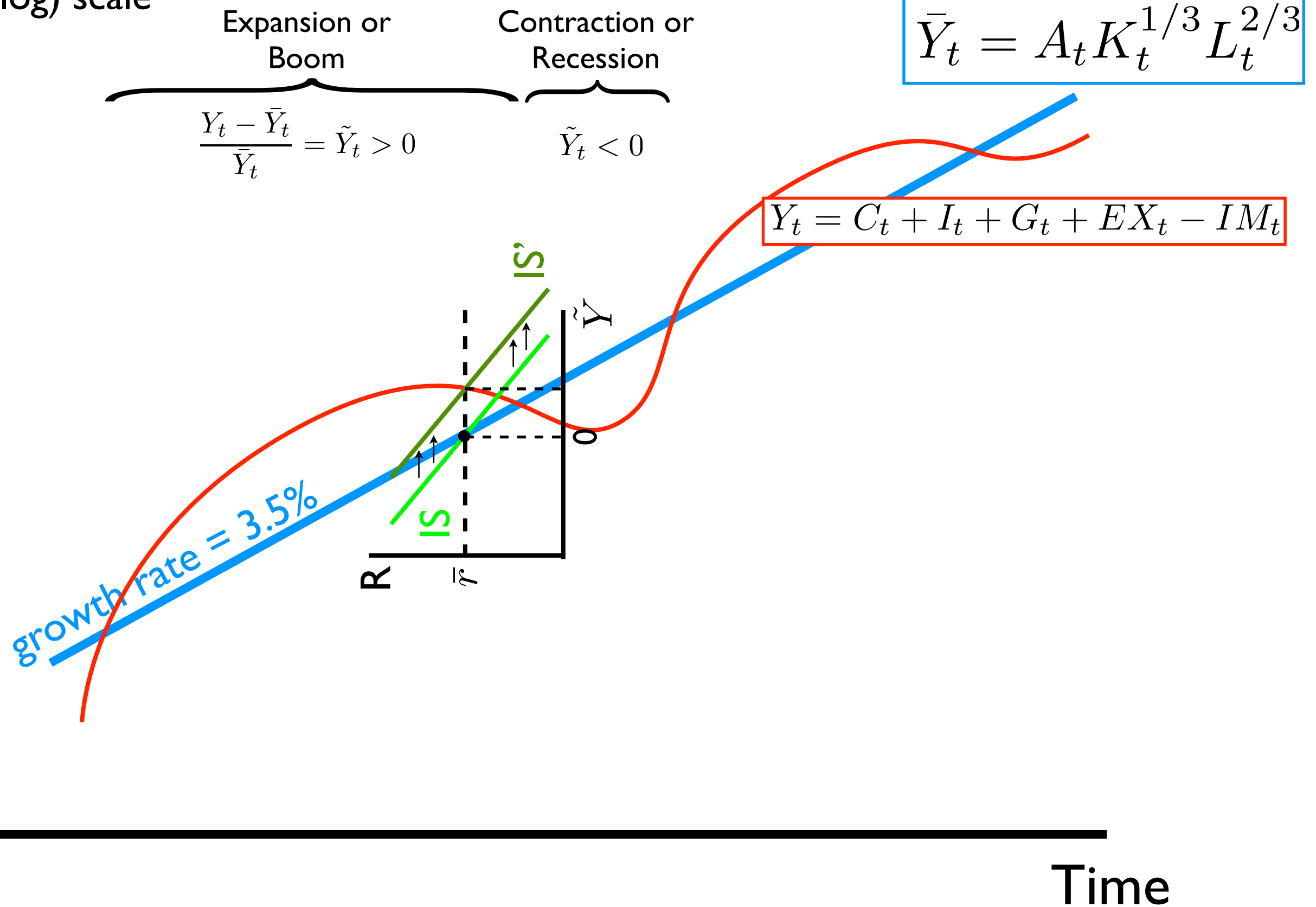
GDP

ratio (log) scale



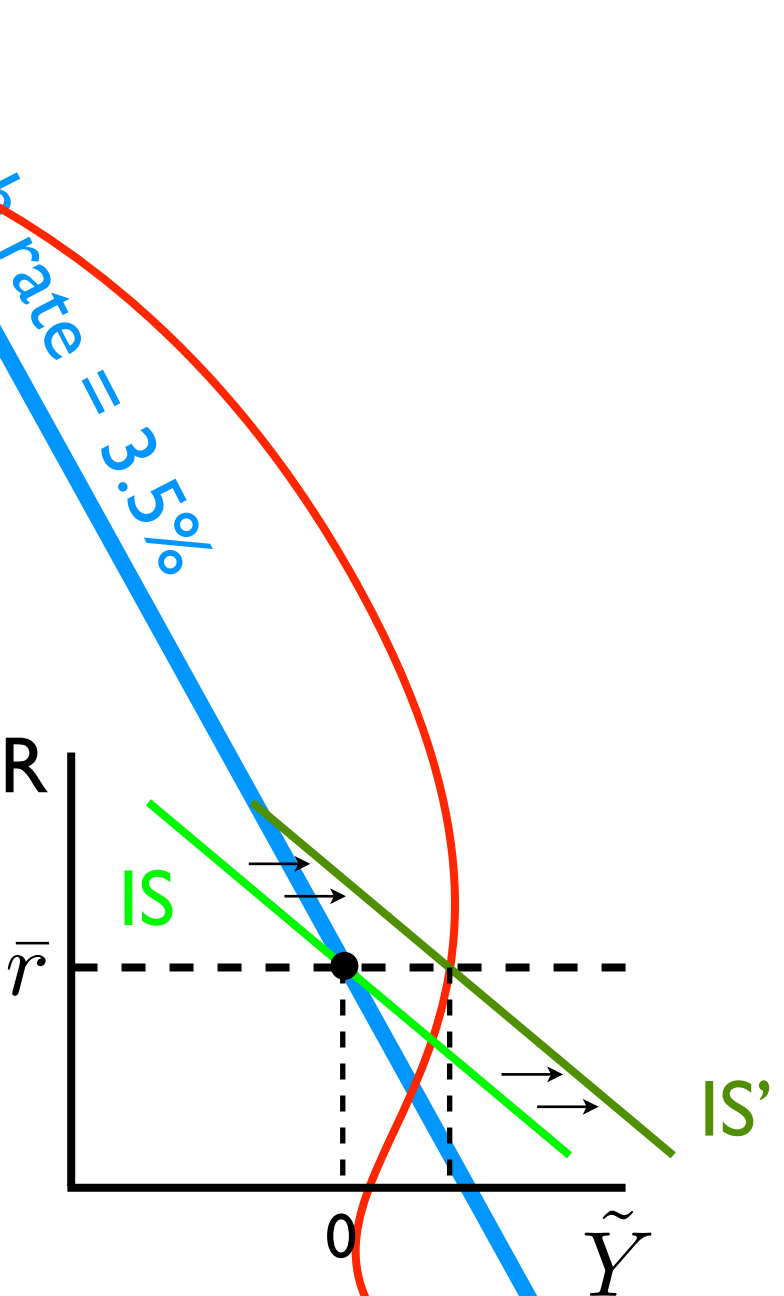
GDP

ratio (log) scale



GDP

ratio (log) scale



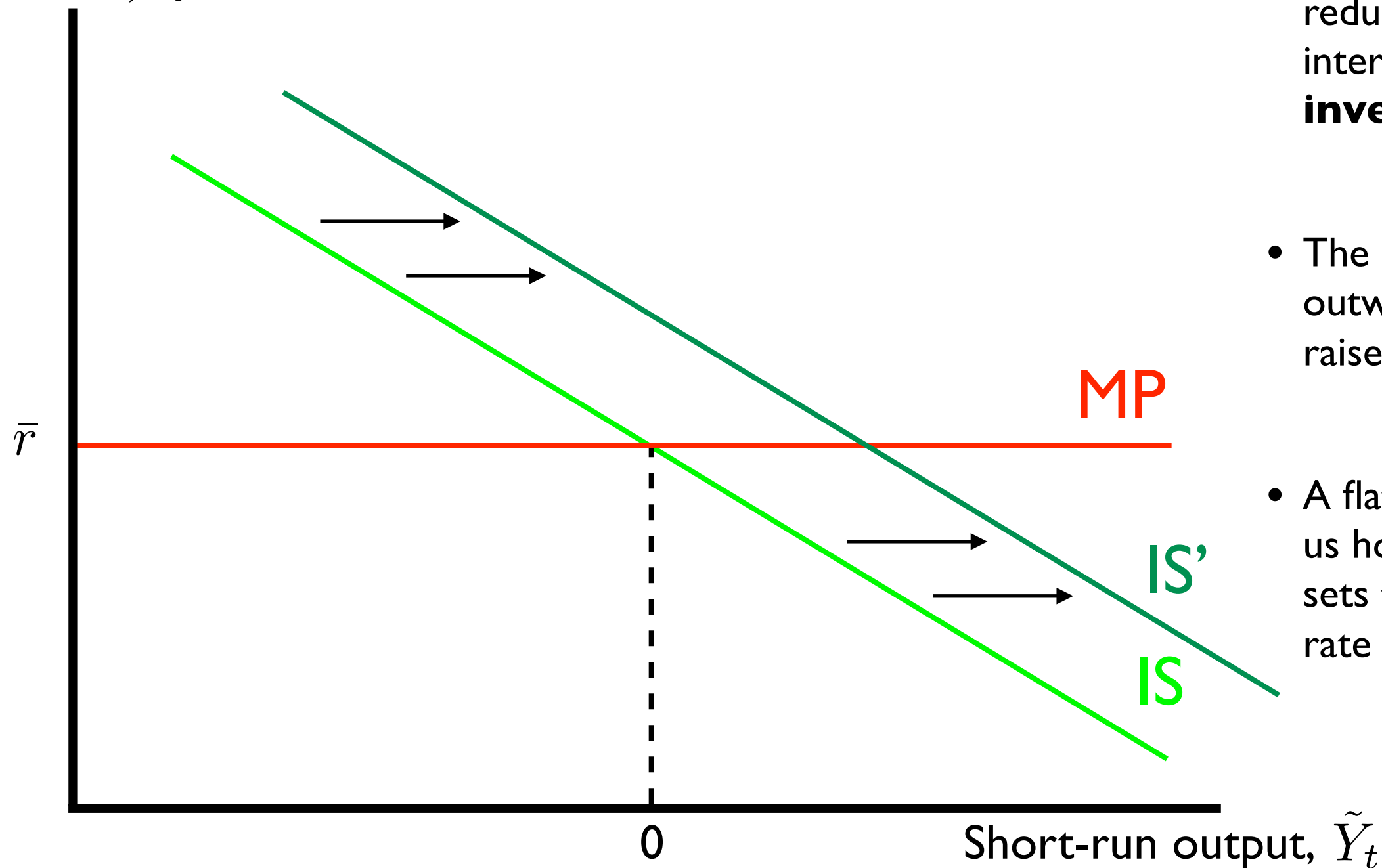
$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

$$\bar{Y}_t = A_t K_t^{1/3} L_t^{2/3}$$

Time

The IS-MP model helps us understand fluctuations

Real interest
rate, R_t



- The IS curve slopes downward because reductions in the real interest rate **raise investment** spending
- The IS curve shifts outward when agents raise spending above $\bar{Y}$
- A flat MP curve shows us how the central bank sets the real interest rate at R ; usually $R = \bar{r}$

What else fluctuates mightily?

Inflation

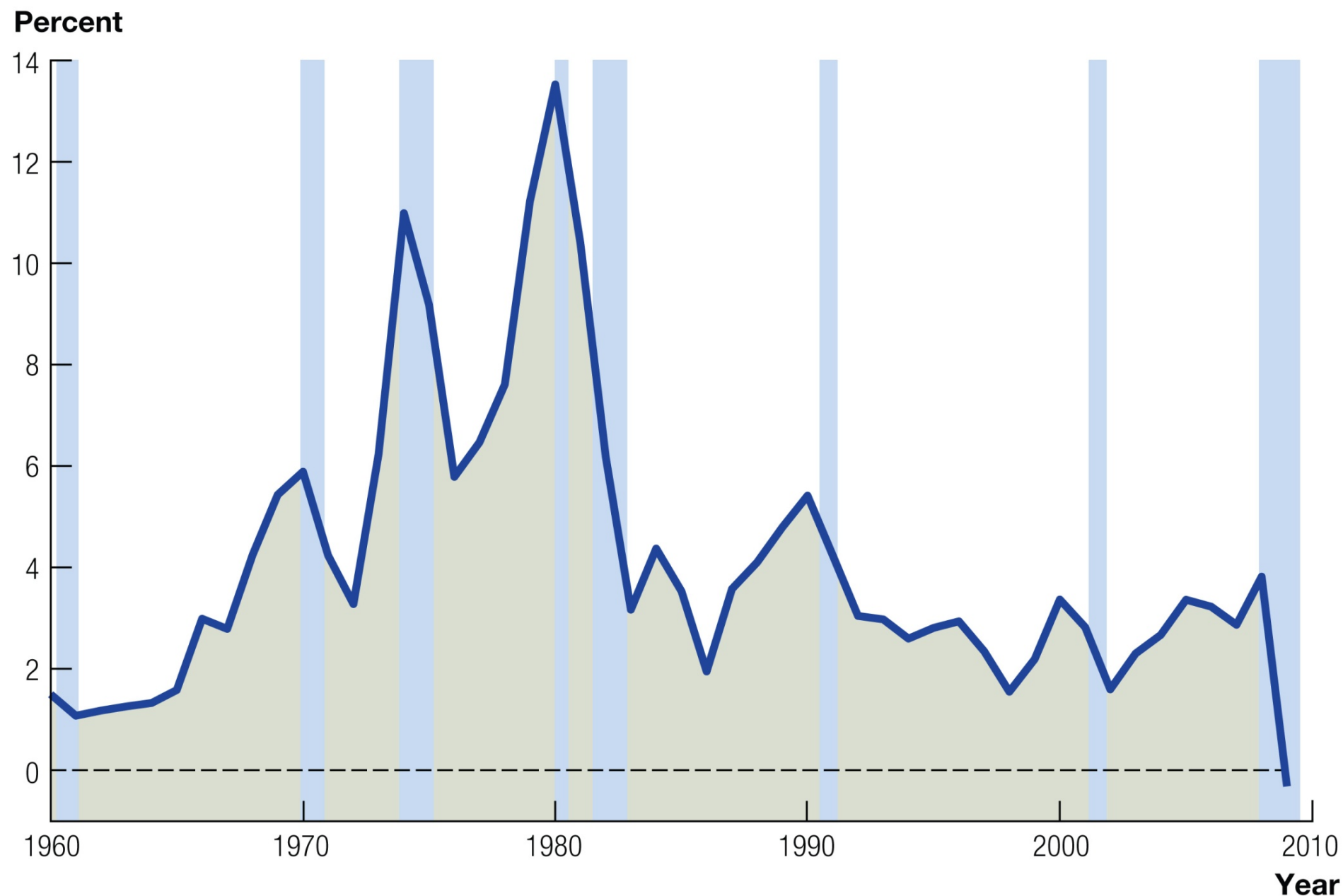
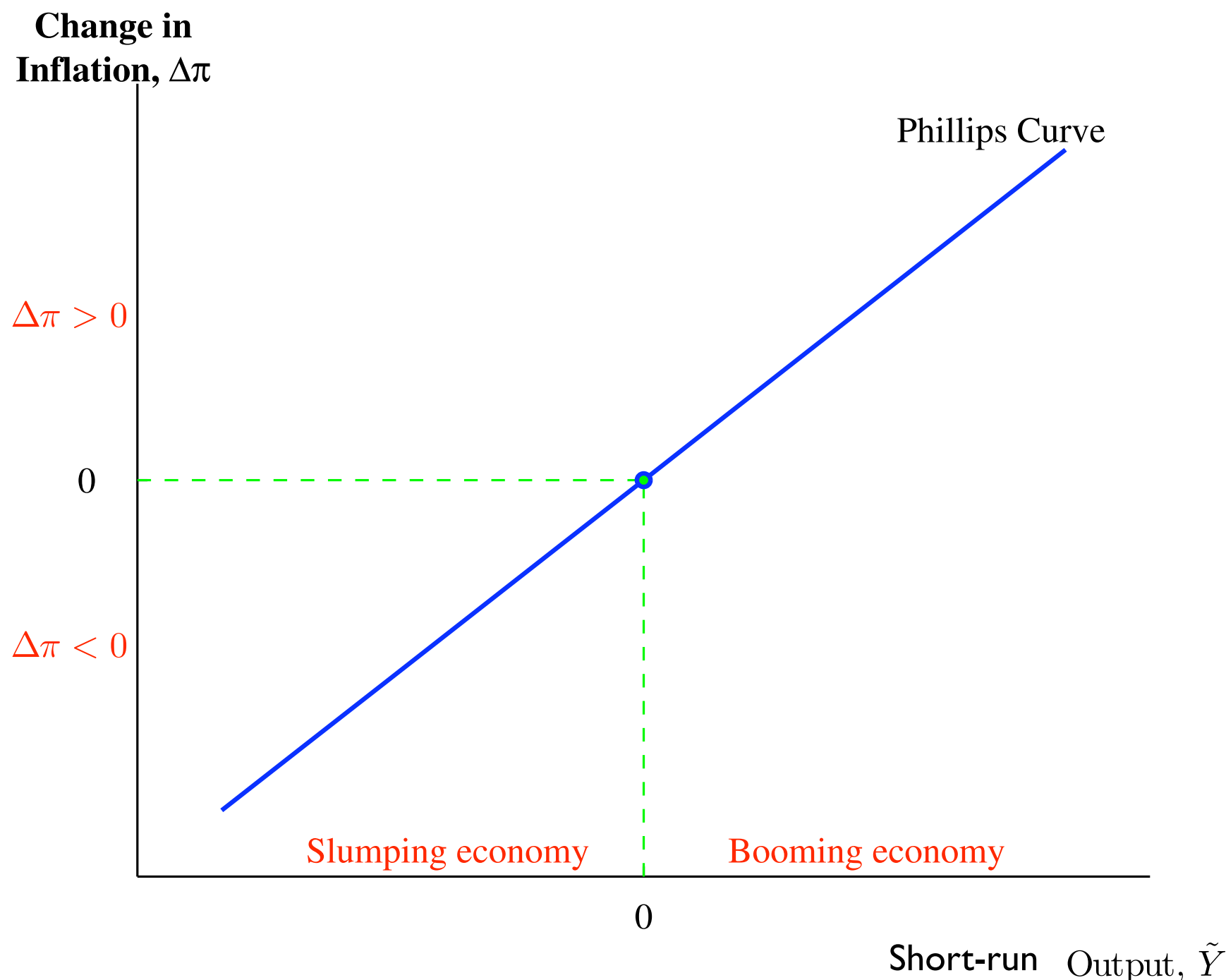


FIGURE 9.5 Inflation in the United States, 1960–2009

Macroeconomics, 2nd Ed
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- This graph shows the **level** of inflation over time, and we see
- During expansions (the white spaces), inflation has risen
- During recessions (the blue spaces), inflation has fallen
- The Phillips Curve shows us how the change in inflation reacts to the business cycle

The Phillips Curve

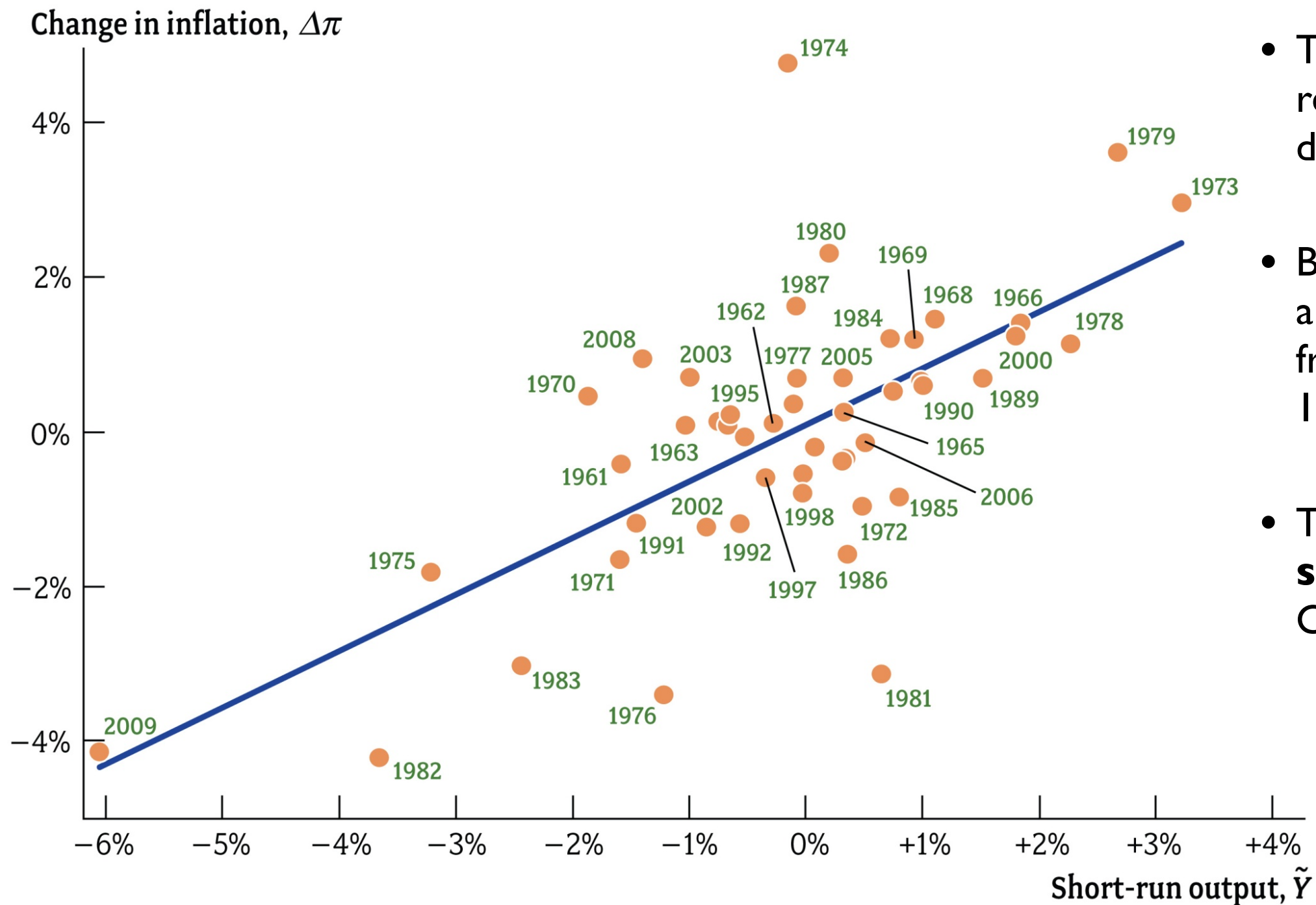


- The relationship between short-run output and inflation is called the **Phillips Curve**
- It is an upward sloping line; if you have output above trend, then you also have an *increase* in inflation
- This is one of the key elements of our short-run model, along with the assumption that policy can change output

Why would inflation rise when output is above trend?

- Think about how output gets above trend
- Suppose there is a shock to demand: it's the holidays, and everybody wants a Furby or Tickle-Me-Elmo or Labubu
- Prices of those goods, at least on eBay, skyrocket
- People have to work overtime to produce extra Elmo's, and they make overtime wages
- Those are good things, but rising prices might not be

Does the Phillips Curve really work?



- The upward sloping relationship is definitely in the data
- But notice that there are many points away from the line, like 1974 and 1981-1982
- There apparently are **shifts** in the Phillips Curve too

FIGURE 9.7 Measuring the Phillips Curve, 1960–2009

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The next several chapters:

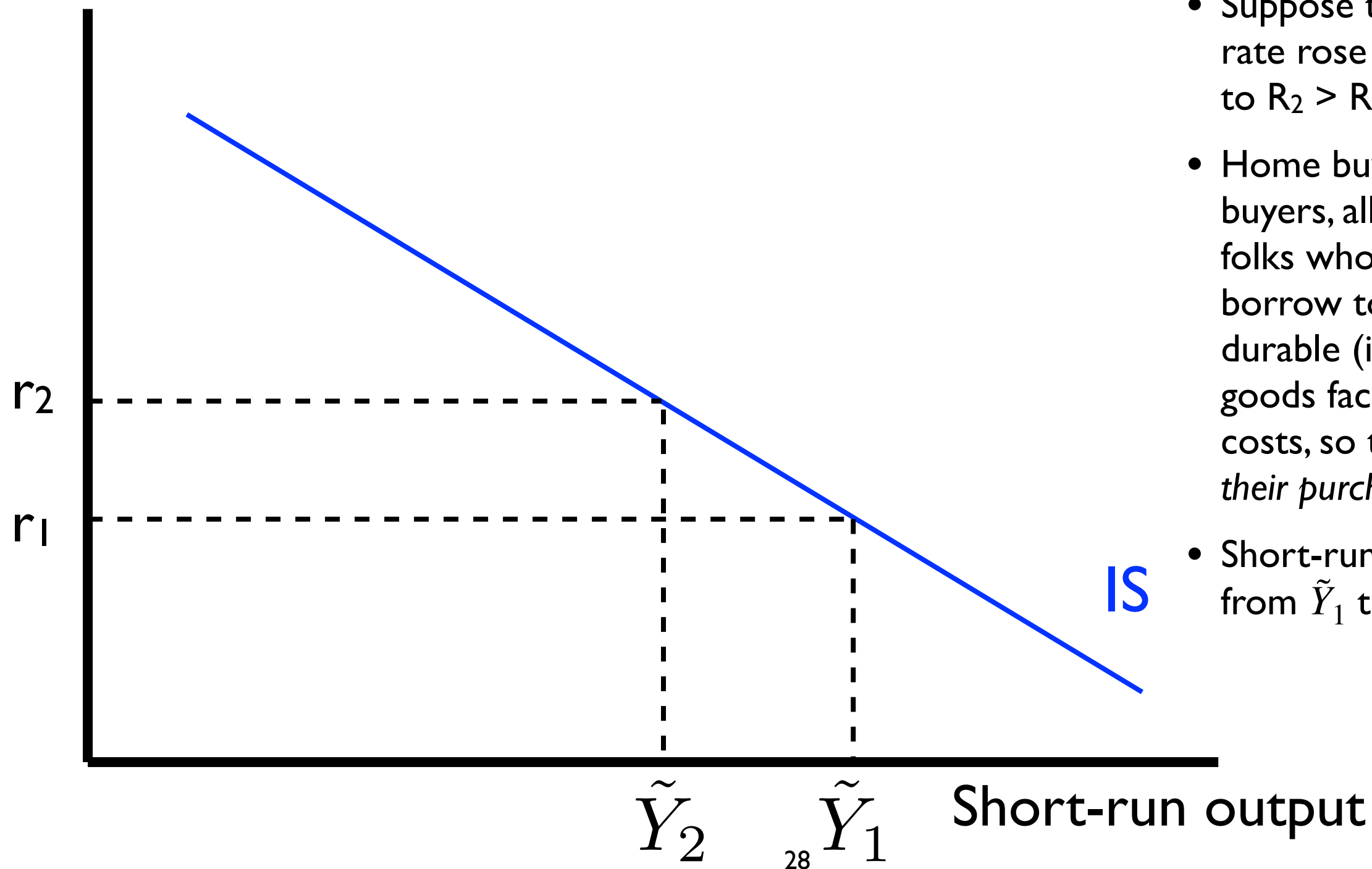
- The “IS Curve” (“eye-ess,” not “izzz”) [Chapter 11]
 - Output in the short run depends negatively on the real interest rate
- Monetary Policy — the “MP” Curve [Chapter 12]
 - Targets the nominal interest rate
- We can combine the IS and MP curves with the Phillips Curve to produce the full Short-Run Model [Chap 13]

A look at the IS curve

- Everybody talks about interest rates all the time
- The Federal Reserve sets interest rates, right?
- Why do we care?
- Why might you care? Are you thinking about:
 - Taking out more student loans? Having to run more credit card debt? Buying a house or a car, for which you'll need to borrow?
 - If yes, then an increase in interest rates would be bad, right?
 - If interest rates rose, you might curtail these activities!
- That's basically the reason why interest rates might affect output in the short run

The IS curve shows activity decreasing when interest rates rise

Interest rate



- Suppose the interest rate rose from R_1 to $R_2 > R_1$
- Home buyers, car buyers, all kinds of folks who have to borrow to buy durable (investment!) goods face higher costs, so they *reduce their purchases*
- Short-run output falls from $\tilde{Y}_1$ to $\tilde{Y}_2$

How do we come up with this?

- We could just specify it, kind of like how a demand curve just always slopes down
- But there's a reason why a demand curve always slopes down: diminishing marginal utility or productivity
- Similarly, there's a reason why the IS Curve slopes down
- Let's begin by reviewing our notation:

Current output we observe

Y_t

Long-run trend or *potential* output

$\bar{Y}_t$

Short-run deviation from long-run potential output

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}.$$

We're going to consider each component of GDP separately

- Why? Each component: Consumption, Investment, Government, Exports, Imports, *behaves a little differently over business cycles*, so we want to treat them separately

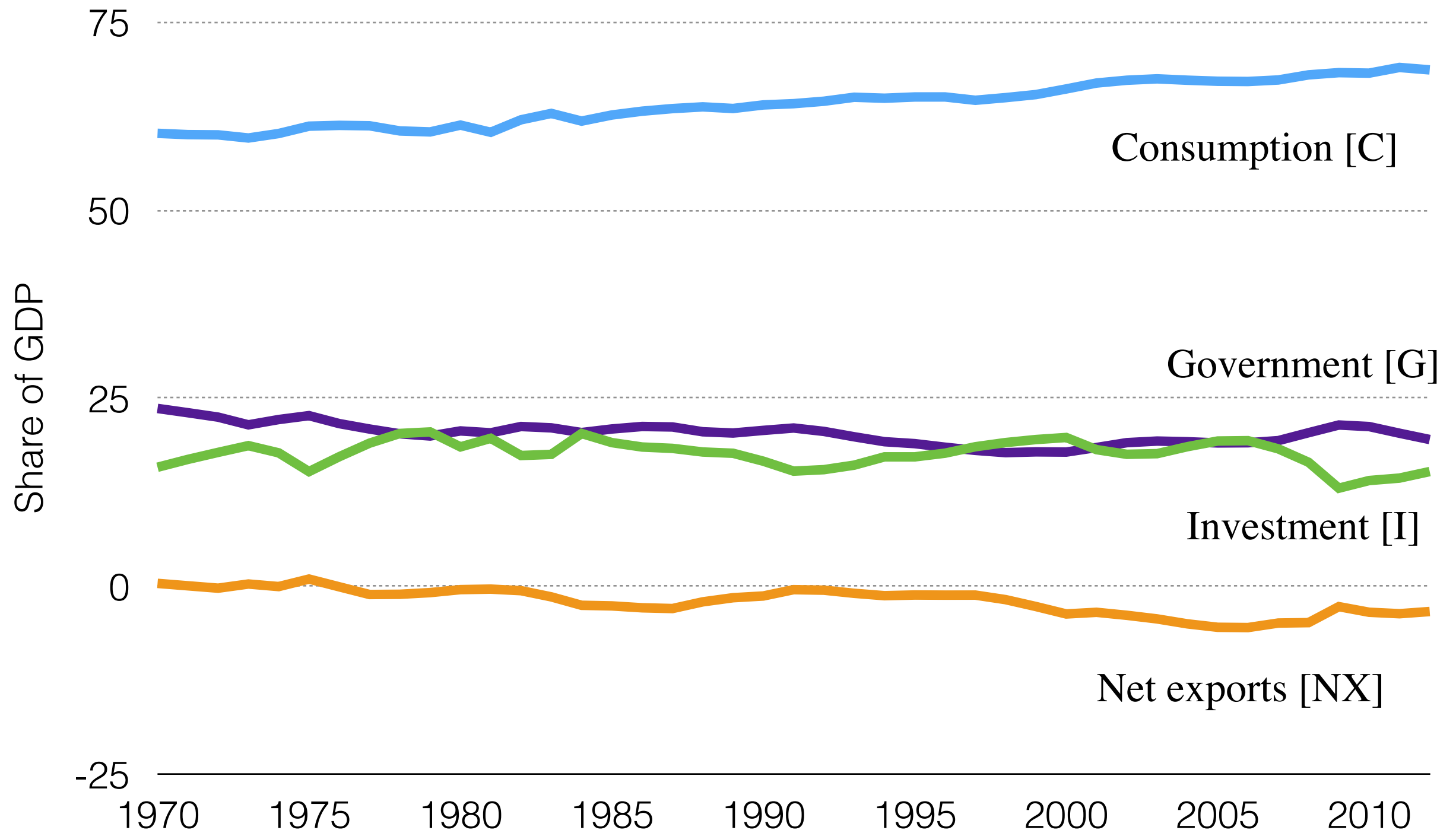
$$Y_t = C_t + I_t + G_t + NX_t$$

- Note that this is Y_t **actual** GDP, *including any fluctuations*
- We're now going to write out **EX**ports and **IM**ports also:

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

- How do each of these behave? We're going to make most of them functions of potential (trend) income, $\bar{Y}_t$

Over time, the expenditure shares of GDP have been stable, with some drift, *except for investment*



Over time, the expenditure shares of GDP have been stable, with some drift, *except for investment*

- As a result, we will model each of the expenditure components *except for investment* as constant shares

- **Consumption** has averaged 64% of GDP since 1970

— Call that $a_c = 0.64$

- **Government** has averaged 20%

— Call that $a_g = 0.20$

- **Exports** have averaged 9%

— Call that $a_{ex} = 0.09$

- **Imports** have averaged 11%

— Call that $a_{im} = 0.11$

Investment has averaged 18% of GDP since 1970, *but it has jumped around a lot too*

so let's set $a_i = 0.18$

and let's model a part of investment that fluctuates around zero

Note that the a 's sum to 1

$$Y_t = \boxed{C_t} + I_t + \boxed{G_t} + \boxed{EX_t} - \boxed{IM_t}$$

$C_t = \bar{a}_c \bar{Y}_t$

$G_t = \bar{a}_g \bar{Y}_t$

$IM_t = \bar{a}_{im} \bar{Y}_t$

Each of these components
is just some fixed share of
potential GDP, $\bar{Y}_t$

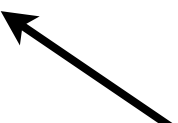
$$EX_t = \bar{a}_{ex} \bar{Y}_t$$

Why do we assume this?

Empirical grounds: Consumption is actually *smoother* than actual output, more like how potential output is smooth over time

Theoretical grounds: If you have savings, then when your income *temporarily* declines, you'll spend some savings to keep consuming

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$



$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b} (R_t - \bar{r})$$

Investment is the special part of all this — it looks a lot different

What does it say? Suppose the right-hand part wasn't there:

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i \longrightarrow I_t = \bar{a}_i \bar{Y}_t \quad \text{Same format as before!}$$

The other part? When the *real interest rate*, R_t , is above the *marginal product of capital*, $\bar{r}$, then $R_t - \bar{r} > 0$ and investment drops **lower**